

Q. How many prime numbers are there from 21 to 30? How many composite numbers are there from 21 to 30?

Ans. Prime numbers from 21 to 30: 2 (23, 29)

Composite numbers from 21 to 30: 8

Q.1. We see that 2 is a prime and also an even number. Is there any other even prime?

Ans. No.

Q.2. Look at the list of primes till 100. What is the smallest difference between two successive primes? What is the largest difference?

Ans. The smallest difference between two successive primes is one ($3 - 2 = 1$).

The largest difference is 8 ($97 - 89$).

Q.3. Are there an equal number of primes occurring in every row in the table on the previous page? Which decades have the least number of primes? Which have the most number of primes?

Ans. No.

The least number of primes occur in the decades 91 to 100.

The most number of primes occur in the decades 1 to 10, 11 to 20.

Q.4. Which of the following numbers are prime: 23, 51, 37, 26?

Ans. $23 = 23 \times 1$

$51 = 3 \times 17 \times 1$

$37 = 37 \times 1$

$26 = 2 \times 13 \times 1$

23 and 37 are prime numbers.

Q.5. Write three pairs of prime numbers less than 20 whose sum is a multiple of 5.

Ans. Pairs of prime numbers less than 20 whose sum is a multiple of 5 are:

(2, 3), (3, 7), (2, 13).

Q.6. The numbers 13 and 31 are prime numbers. Both these numbers have same digits 1 and 3. Find such pairs of prime numbers up to 100.

Ans. 17 and 71, 37 and 73, 79 and 97.

Q.7. Find seven consecutive composite numbers between 1 and 100.

Ans. 90, 91, 92, 93, 94, 95, 96.

Q.8. Twin primes are pairs of primes having a difference of 2. For example, 3 and 5 are twin primes. So are 17 and 19. Find the other twin primes between 1 and 100.

Ans. 3 and 5,

5 and 7,

11 and 13,

17 and 19,

29 and 31,

41 and 43,

59 and 61,

71 and 73.

Q.9. Identify whether each statement is true or false. Explain.

Statements:

- a. There is no prime number whose units digit is 4.
- b. A product of primes can also be prime.
- c. Prime numbers do not have any factors.
- d. All even numbers are composite numbers.
- e. 2 is a prime and so is the next number, 3. For every other prime, the next number is composite.

Ans.

- a. True, unit digit 4 means even number. We know that 2 is the only even prime number.
- b. False, product of primes becomes a composite number.
- c. False, prime numbers have exactly two factors: 1 and the number itself.
- d. False, all even numbers except 2 are composite numbers.
- e. True, Except 2, for every prime number, the next number is composite.

Q.10. Which of the following numbers is the product of exactly three distinct prime numbers: 45, 60, 91, 105, 330?

Ans. $45 = 3 \times 3 \times 5$

$$60 = 2 \times 2 \times 3 \times 5$$

$$91 = 7 \times 13$$

$$105 = 3 \times 5 \times 7$$

$$330 = 2 \times 3 \times 5 \times 11$$

105 is the only number which is the product of three distinct prime numbers.

Q.11. How many three-digit prime numbers can you make using each of 2, 4 and 5 once?

Ans. There is no prime number that can be made using the digit 2, 4, and 5 because if the unit place is 2 or 4, then it is divisible by 2, and if the unit place is 5, then it is divisible by 5.

Q.12. Observe that 3 is a prime number, and $2 \times 3 + 1 = 7$ is also a prime. Are there other primes for which doubling and adding 1 gives another prime? Find at least five such examples.

Ans. Yes,

- $11 = 2 \times 5 + 1$
- $47 = 2 \times 23 + 1$
- $83 = 2 \times 41 + 1$
- $23 = 2 \times 11 + 1$
- $59 = 2 \times 29 + 1$